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RATIO AND PROPORTION, VARIATION

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Question 1

If a : b = 7 : 9, b : c = 3 : 5, c : d = 3 : 7 and d is 25% more than e, find the ratio between a and e.

A) 3/4

B) 1/4

C) 4/3

D) 4/5

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Question 2

What number must be subtracted from each of the numbers 47, 19, 56, 22 so that the remainders are in proportion?

A) 9

B) 5

C) 7

D) 3

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Question 3

There are some apples and oranges in the ratio 3 : 14. If five apples are eaten the ratio will become 1 : 8. Find the number of oranges.

A) 56

B) 48

C) 52

D) 46

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Question 4

Rs. 3456 is divided among two men, a woman and five boys. The ratio of share of a man, a woman and a boy is 8 : 5 : 3. What is the share of a boy?

A) Rs. 288

B) Rs. 480

C) Rs. 576

D) Rs. 460.8

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Question 5

Two equal containers are filled with the mixture of milk and water. The concentration of milk in each of the containers is 24% and 36%, respectively. What is the ratio of water in both the containers respectively?

A) 7:3

B) 21:17

C) 19:16

D) 2:3

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Question 6

A box contains 25 paise, 50 paise and Re. 1 coins in the ratio 4 : 9 : 7. What is the number of coins of 50 paise if the difference between the amount yielded by 25 paise and Re. 1 coin is Rs. 24?

A) 36

B) 16

C) 24

D) 70

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Question 7

The workers in three shifts of a company are in the ratio of 2 : 3 : 4. If 64 workers are added in each shift, the ratio becomes 4 : 5 : 6. Find the total number of workers in all the three shifts.

A) 480

B) 288

C) 352

D) 576

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Question 8

The value of a coin varies directly with the square of its radius when its thickness is constant. The radius of a coin is 2 cm and its value is Rs. 5. What will be the radius of a coin if its value is Rs. 10?

A) 2.83 cm

B) 5 cm

C) 3.23 cm

D) 2.5 cm

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Question 9

The ratio of students in a coaching preparing for JEE and CAT is 2 : 3. The ratio of fees collected from each of JEE and CAT student is 9 : 4. If the total amount collected from all the students is 1.70 lakhs, what is the total amount collected from only CAT aspirants?

A) 1.02 lakhs

B) 0.68 lakh

C) 0.6 lakh

D) 1 lakh

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Question 10

Two distinct numbers are in the ratio X : Y. When 3 is added to the numerator and 1 is added to the denominator, the ratio gets changed to 2 : 1. The sum of X and Y could be _____.

A) 4

B) 5

C) 6

D) 7

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Question 11

Krishna has three dogs and they are roaming in such a way that when dog A takes 6 steps, dog B takes 7 steps and dog C takes 8 steps. But the 7 steps of A are equal to the 8 steps of B and 9 steps of C. What is the ratio of their speeds?

A) 432 : 441 : 448

B) 448 : 441 : 432

C) 196 : 192 : 189

D) 196 : 192 : 189

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Question 12

Four balls are numbered and dropped in a box. The four numbers are in proportion. The sum of the squares is 50 and the sum of the means is 5. The ratio of first two is 1 : 3. What is the average of the four numbers on the balls?

A) 12

B) 3

C) 6

D) 4

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Question 13

Indian Army stored guns at an Arsenal A. Every month 40 new recruits joined the army and took two guns each. After 5 months, the number of new recruits per month decreased by 10. The guns lasted for 45 months. If the ratio of guns in Arsenal A to Arsenal B is 7 : 10, find the number of guns in Arsenal B.

A) 4000

B) 3600

C) 4200

D) 3200

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Question 14

Three containers contain mixtures of alcohol and apple juice in the ratio 5 : 11, 3 : 2 and 7 : 17. If all three of them are mixed together in the ratio 4 : 5 : 6, find the ratio of alcohol to apple juice in the final mixture.

A) 1/3

B) 2/3

C) 3/4

D) 4/5

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Question 15

Three containers contain mixtures of alcohol and apple juice in the ratio 5 : 11, 3 : 2 and 7 : 17. If all three of them are mixed together in the ratio 4 : 5 : 6, how much pure alcohol (in ml) must be added to 1 litre of the final mixture to get a stress relieving medicine which has 60% pure alcohol in it?

A) 450

B) 500

C) 550

D) 600

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Question 16

If A is directly proportional to B and C and inversely proportional to D, find the change in B when A, C and D vary by factors of 14/9,8/3,6/7, respectively.

A) 2/3

B) 1/2

C) 1/3

D) 3/2

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Question 17

Three friends A, B and C invested a total of Rs. 500000 in the shares of FACE. FACE shares compounded by 20% for 3 years. If the profits received by A and B are Rs. 91000 and Rs. 113750, calculate the amount invested by A.

A) 128950

B) 120000

C) 125000

D) 150000

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Question 18

A milkman has a mixture of milk in which ratio of milk and water is 7 : 4. He sells 55 litres of milk and then he adds up 20 litres of pure water. Now, the ratio of milk and water is 7 : 6. What is the new quantity of mixture?

A) 165

B) 15

C) 130

D) 185

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Question 19

The value of a diamond is directly proportional to the square of its weight. A diamond unfortunately breaks into three pieces with weights in the ratio 5 : 6 : 7 and thus, a loss of Rs. 21.4 lakhs is incurred. What is the actual value of the diamond?

A) 42.8 lakhs

B) 31.9 lakhs

C) 32.4 lakhs

D) 39.4 lakhs

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Question 20

A tin contains a mixture of water and alcohol in the ratio of 7 : 3 and another tin contains the water and alcohol in the ratio of 5 : 4. In what proportion should the solution of two tins be mixed to achieve a perfect proportion of 2 : 1 (in which water is two times that of alcohol).

A) 9:4

B) 3:10

C) 4:9

D) 10:3

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THANK YOU

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